

Constrained DAG Structure Learning via Variational Auto Encoder

Group member:
Major:

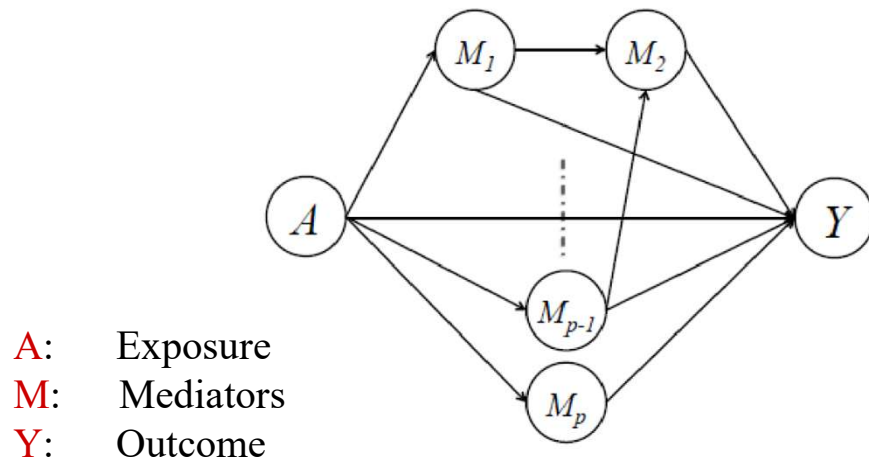
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Bioinformatics

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Bioinformatics

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Computer Science

ECE 542: Neural Network

1.Object



A causal graph with interacted mediators.

$$\mathcal{G} = (X, \mathbf{B}) \quad X = [A, M^T, Y]^T$$

	A	M_1	M_2	$\dots$	M_p	Y
A	0	0	0	0	0	0
M_1	α_1	0				0
M_2	α_2		0	$\mathbf{b}_{ij}$		0
$\vdots$	α_i		$\mathbf{b}_{ij}$	0		0
M_p	α_p				0	0
Y	γ	β_1	β_2	β_j	β_p	0

Weighted DAG with weight matrix $\mathbf{B}$.

Figure 1: The illustration of **DAGs with different types of mediators**.

Left: Network representation

Right: Graph weighted matrix representation

2.Method

2.1 Linear structural equation model(LSEM)

$$X = B^T X + \varepsilon$$

$$\mathcal{G} = (X, B)$$

$$X = [A, M^T, Y]^T$$

Weighted DAG with weight matrix B.



$$\begin{bmatrix} A \\ M^T \\ Y \end{bmatrix} = B^T \begin{bmatrix} A \\ M^T \\ Y \end{bmatrix} + \varepsilon = \begin{pmatrix} 0 & 0_{p \times 1} & 0 \\ \alpha & B_M^T & 0 \\ \gamma & \beta^T & 0 \end{pmatrix} \begin{bmatrix} A \\ M^T \\ Y \end{bmatrix} + \begin{bmatrix} \varepsilon_A \\ \varepsilon_{M_p} \\ \varepsilon_Y \end{bmatrix}$$



$$A = \varepsilon_A$$

$$M^T = (I - B_M^T)^{-1} (\alpha A + \varepsilon_{M_p})$$

$$Y = \gamma A + \beta^T M^T + \varepsilon_Y$$

2.2 Constrained structural learning—ANOCE: CVAE algorithm

A type of deep generative model, [Variational Autoencoders: VAE](#)

– assume actual data is generated from probability distribution of some latent variable

$$X = B^T X + \varepsilon \quad \Rightarrow \quad X = (I_{p+2} - B^T)^{-1} \varepsilon$$

Random error ε treated as the independent latent variables to generate X

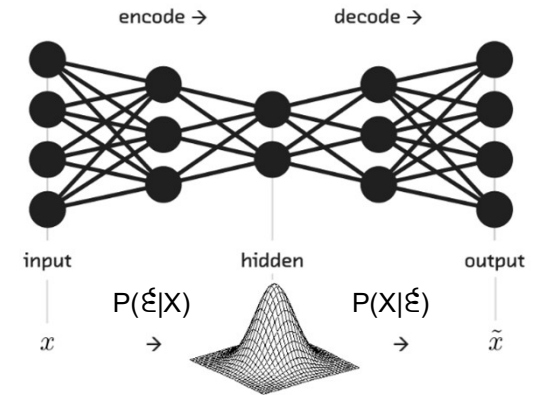


Figure 2: VAE model

Objective function and penalties:

Evidence Lower Bound (ELBO)

$$\min_{B, \theta} f(B, \theta) = \frac{1}{p+2} \sum_{i=1}^{p+2} D_{KL}\{q(\epsilon|X_i) || p(\epsilon)\} - E_{q(\epsilon|X_i)}\{\log p(X_i|\epsilon)\}$$

$$h_1(B) \equiv \text{tr}[(I_{p+2} + tB \bullet B)^{p+2}] - (p+2) = 0.$$

Acyclicity constraint

$$h_2(B) \equiv \sum_{i=1}^{p+2} |b_{i,1}| + \sum_{j=2}^{p+2} |b_{p+2,j}| = 0$$

Topological ordering constraint

3.Data

Branch 1: Real Data Set

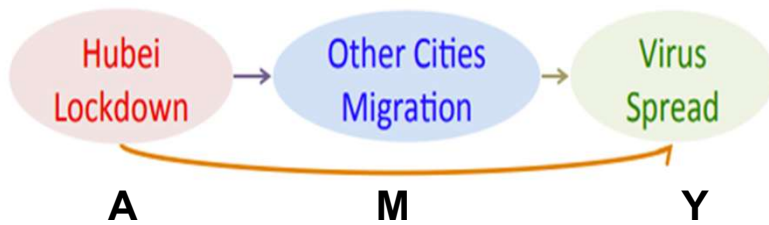


Figure 3: Real data structure

Branch 2: Simulation Data Set

Generated true DAG with interacted mediators from a random procedure.

Number of mediators $p = 10$.

Evaluation metrics: TPR, FPR, F1-score, **S**tructural **H**amming **D**istance

- Exposure A (1 exposure): Locking Wuhan down on Jan 23rd, 2020
- Mediators M (30 mediators): Migration scale of major Chinese cities outside Hubei
Significantly reduced
- Outcome Y (1 outcome): Stopping virus spreading
- Sample size: 38

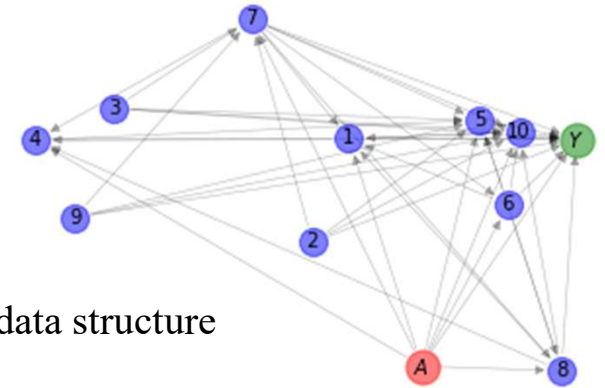


Figure 4: Simulation data structure

SHD, the smallest number of edge additions, deletions, and reversals to convert the estimated graph into the true DAG

4.Results Real data: Causal effect of “Wuhan lockdown” on virus spreading

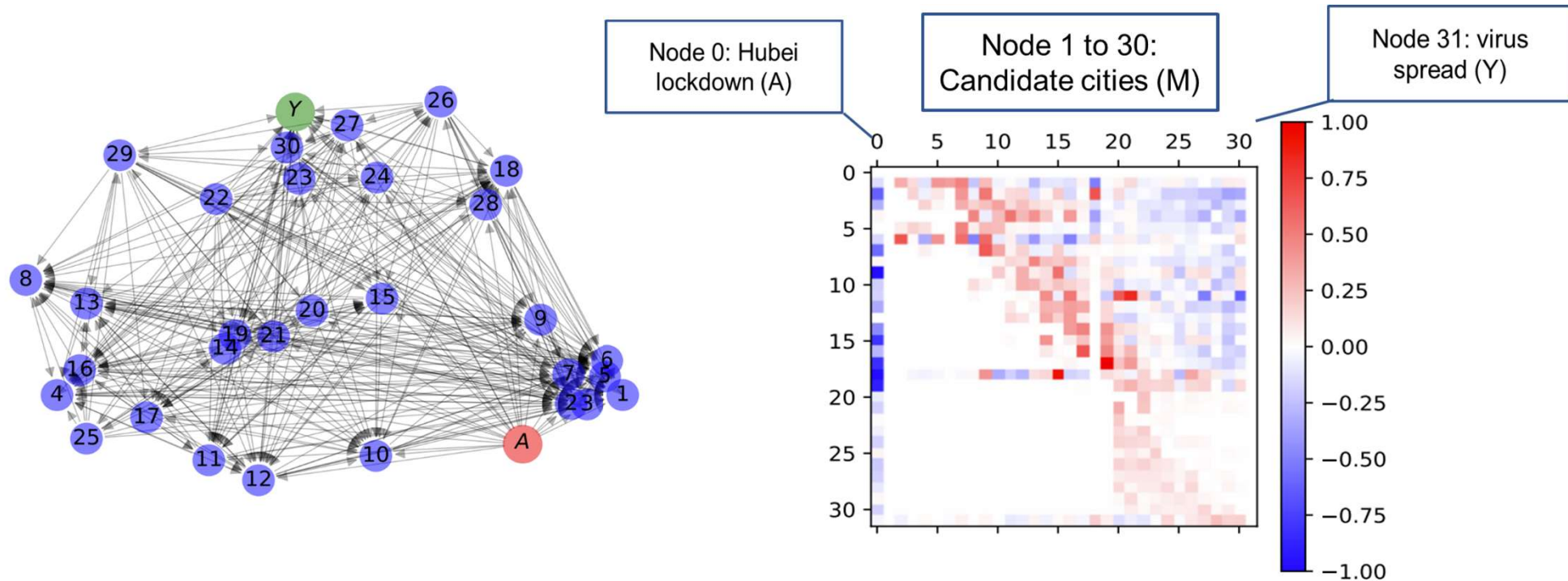
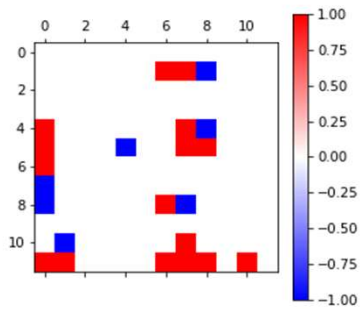


Figure 5: Causal effect of “Wuhan lockdown” on virus spreading

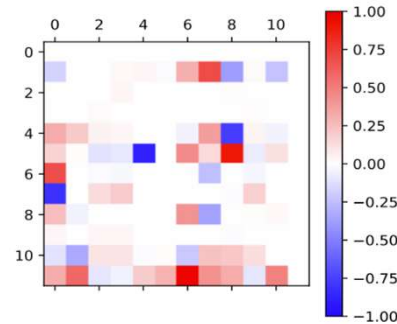
Left : Network representation

Right: Graph weighted matrix representation

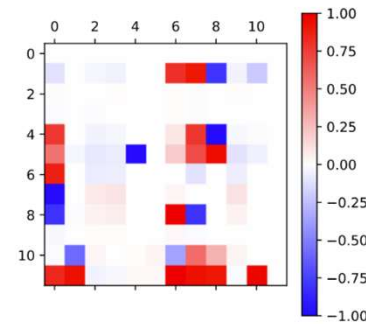
Simulation: Increase Number of Sample Size



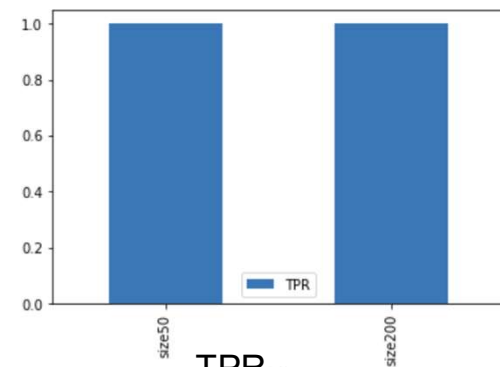
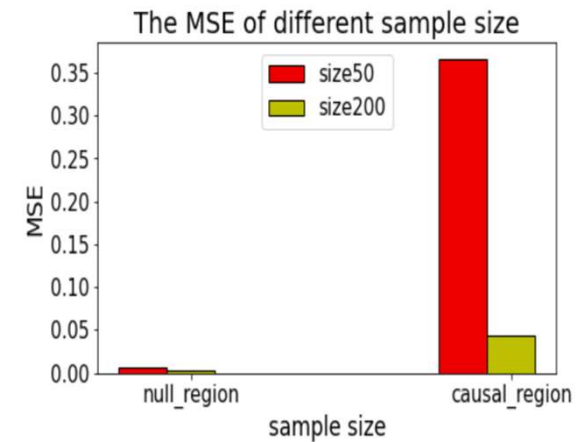
True



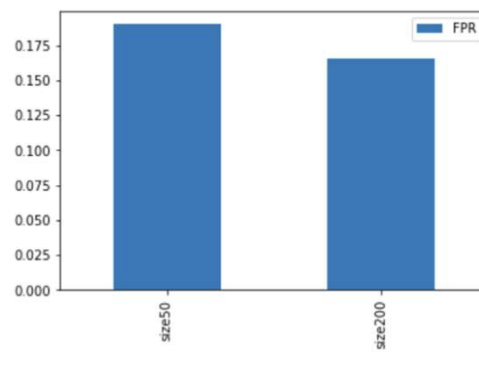
Sample Size=50
Baseline setting



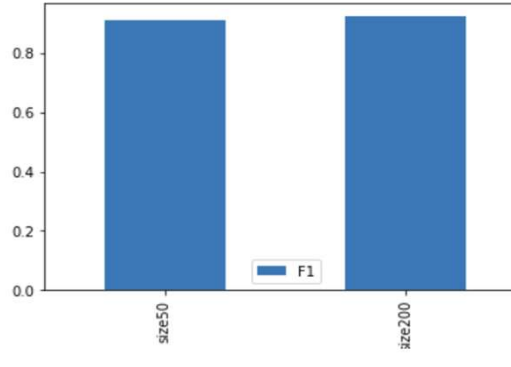
Sample Size=200



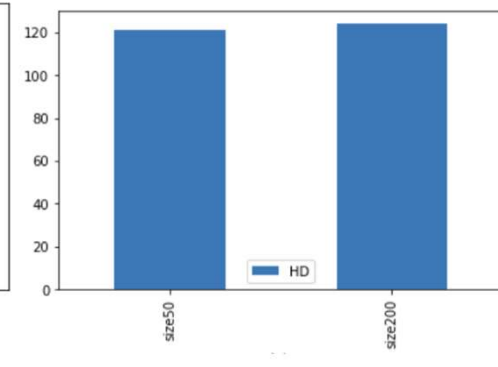
TPR



FPR



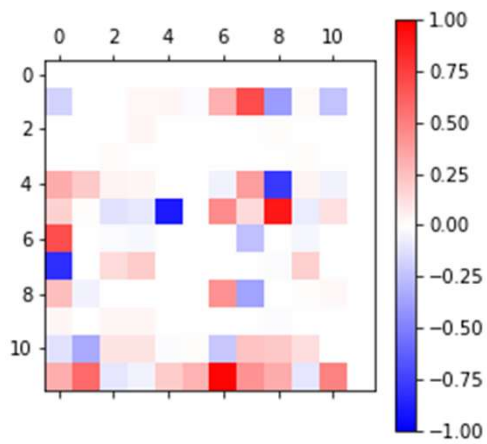
F1-score



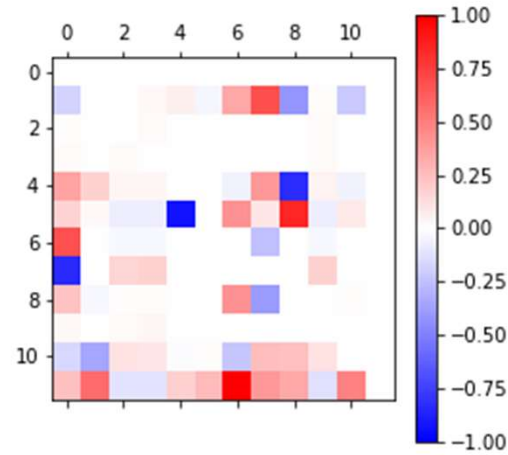
Structural Hamming distance

SHD, the smallest number of edge additions, deletions, and reversals to convert the estimated graph into the true DAG

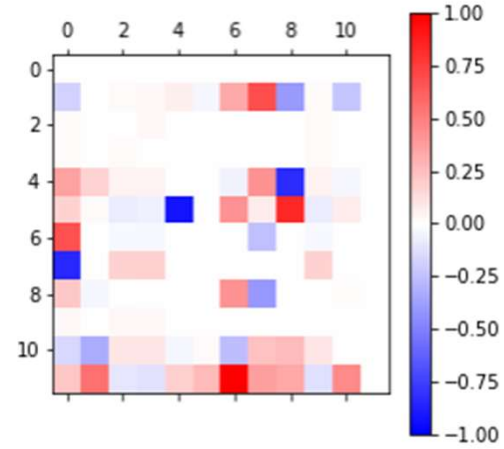
Simulation: random sampling replications



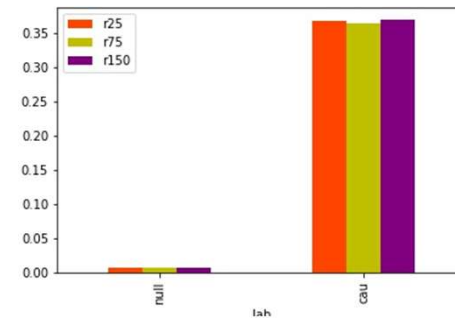
Rep=25
Baseline setting



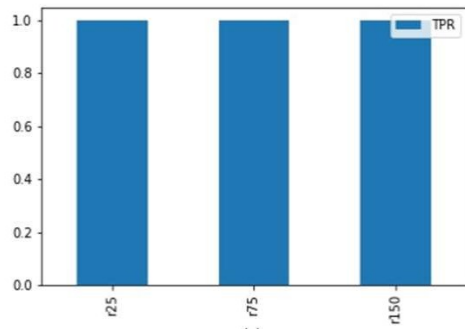
Rep=75



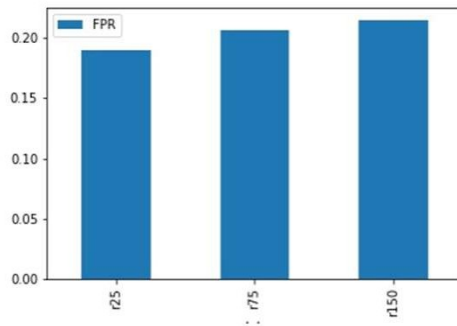
Rep=150



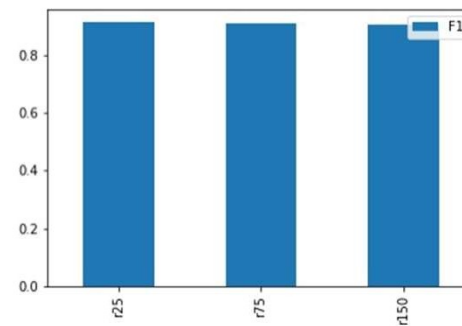
MSE



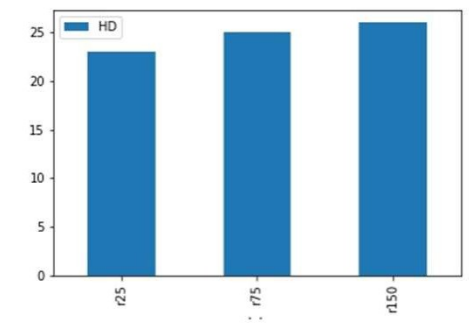
TPR



FPR



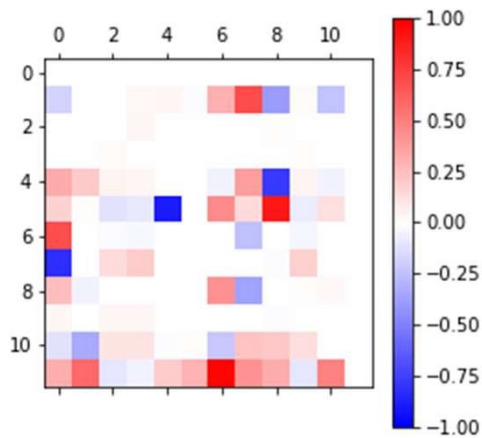
F1-score



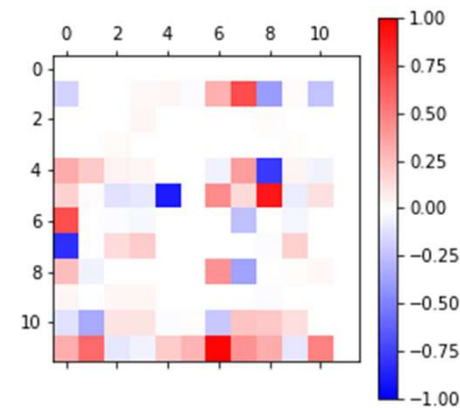
Structural Hamming distance

SHD, the smallest number of edge additions, deletions, and reversals to convert the estimated graph into the true DAG

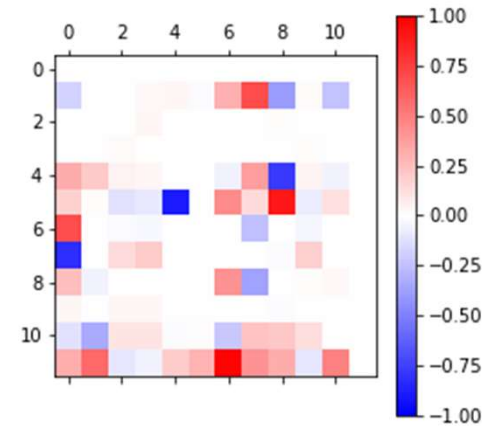
simulation: Learning rate decay



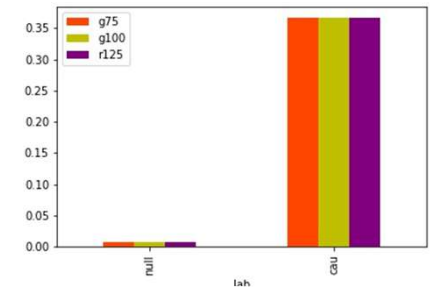
Learning rate decay 1
Baseline setting



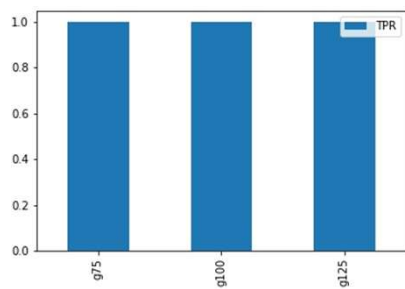
Learning rate decay 0.75



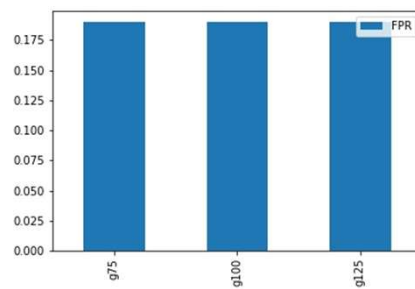
Learning rate decay 1.25



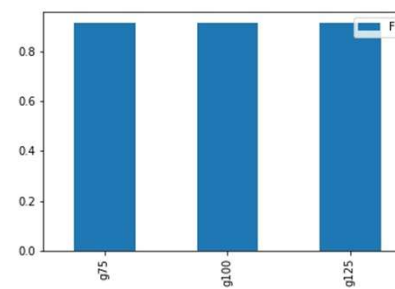
MSE



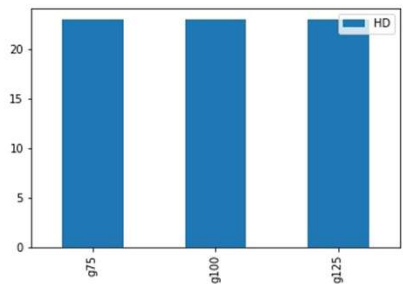
TPR



FPR



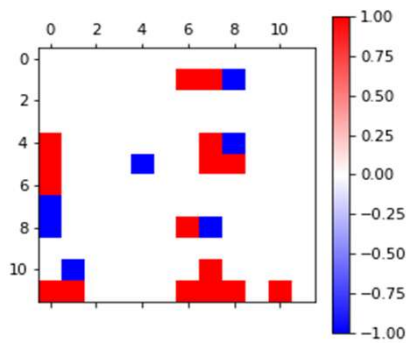
F1-score



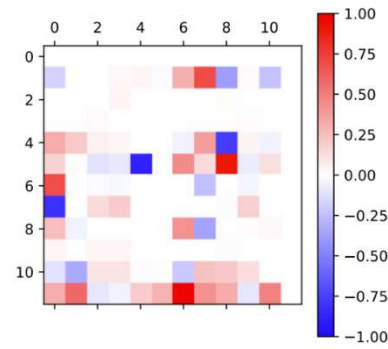
Structural Hamming distance

SHD, the smallest number of edge additions, deletions, and reversals to convert the estimated graph into the true DAG

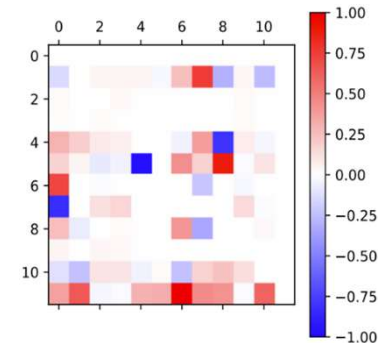
simulation: increase number of hidden layers



True

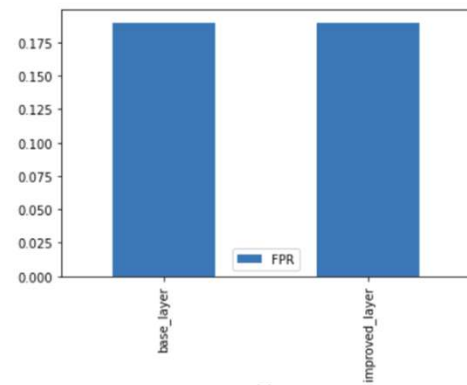


Baseline Hidden Layer

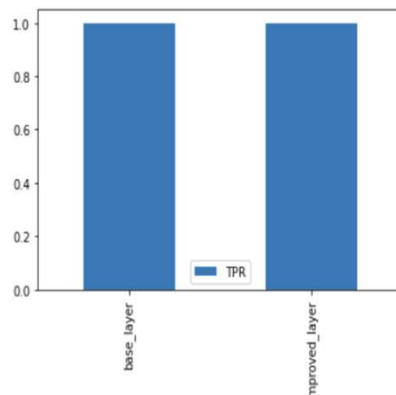


Improved Hidden Layer

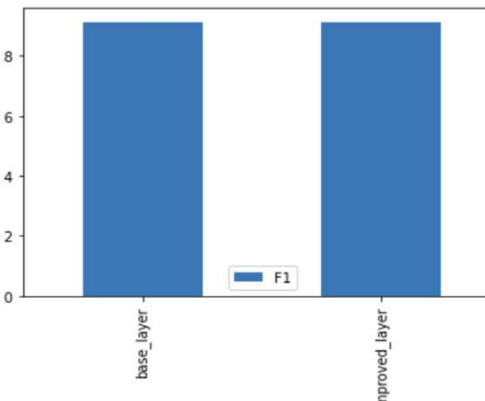
Improvement of Hidden Layer Structure



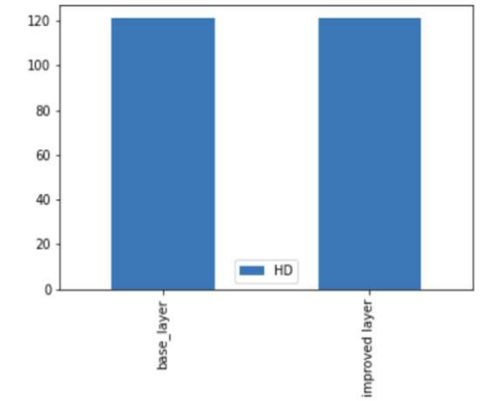
FPR



TPR



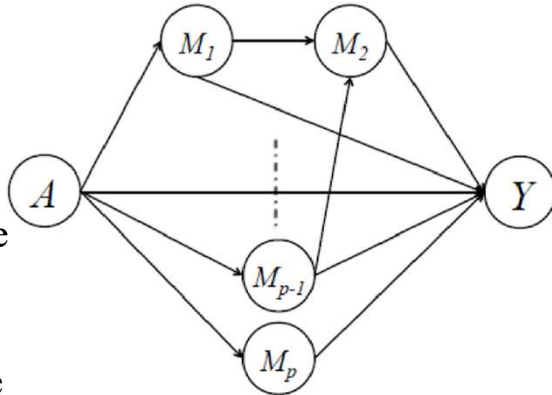
F1



HD

Summary

1. Object



A : Exposure

M : Mediators

Y : Outcome

2. Method

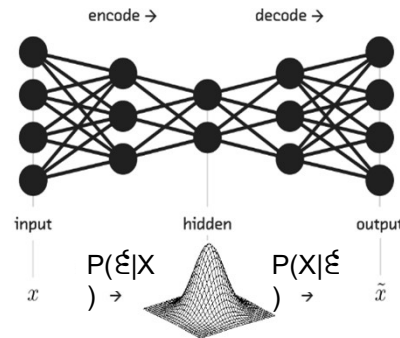
$$\mathcal{G} = (X, B) \quad X = [A, M^T, Y]^T$$

$$X = (I_{p+2} - B^T)^{-1} \varepsilon$$

$$A = \varepsilon_A$$

$$M^T = (I - B_M^T)^{-1} (\alpha A + \varepsilon_{M_p})$$

$$Y = \gamma A + \beta^T M^T + \varepsilon_Y$$

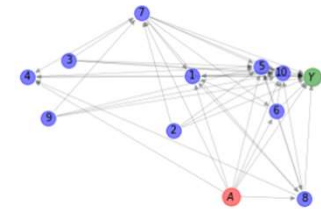


3. Data

3.1 Real Data



3.2 Sim Data

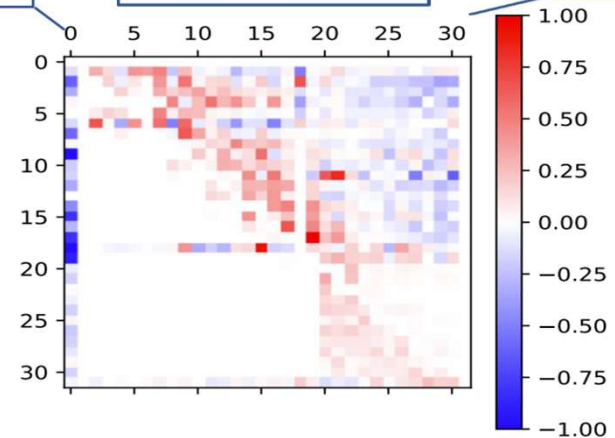


4. Results

Node 0: Hubei lockdown (A)

Node 1 to 30: Candidate cities (M)

Node 31: virus spread (Y)



Thank you!

VAE Constraints

Where $h_1(B)$ denotes the acyclicity constraint;

$$h_1(B) \equiv \text{tr}[(I_{p+2} + tB \bullet B)^{p+2}] - (p+2) = 0$$

And $h_2(B)$ denotes the background knowledge: Temporal order A as the first order, Y as the last one;

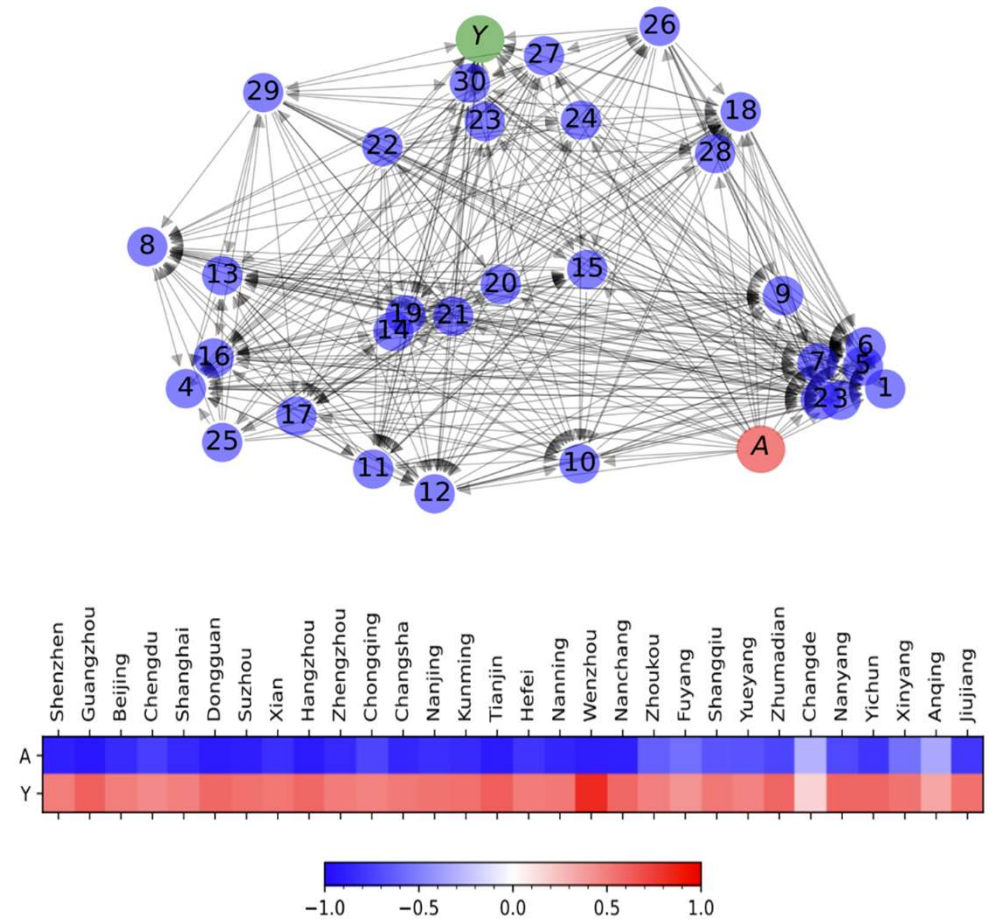
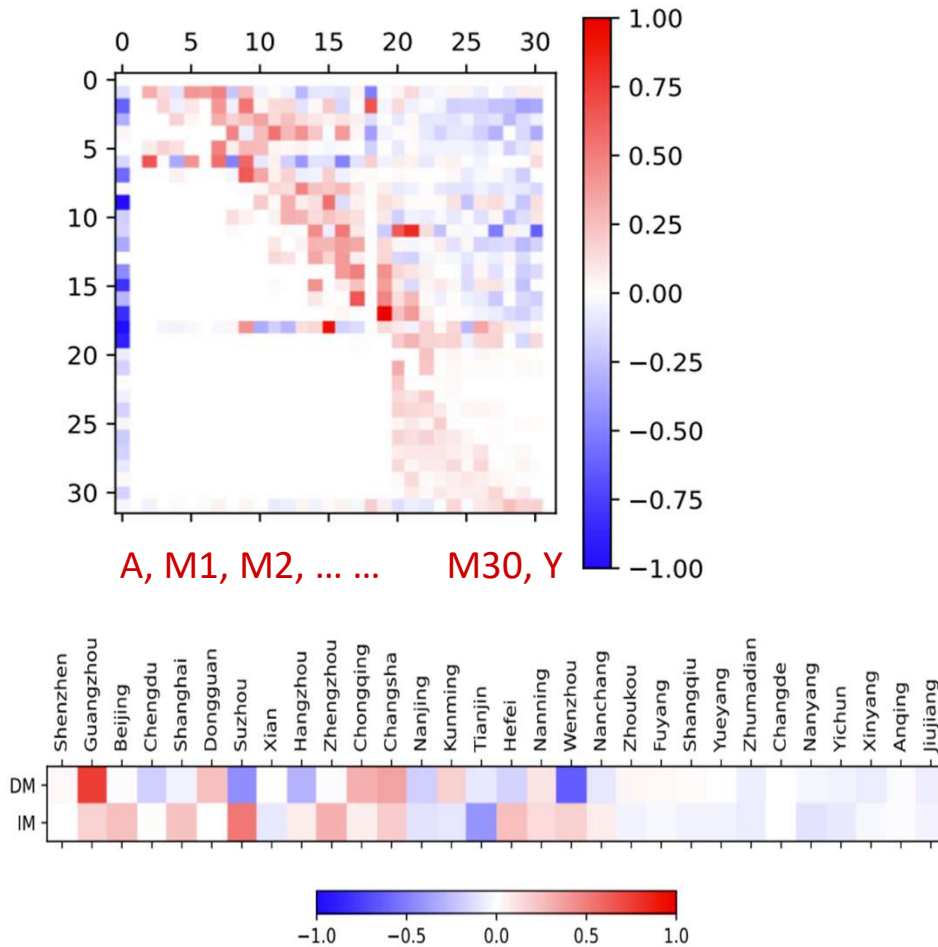
$$h_2(B) \equiv \sum_{i=1}^{p+2} |b_{i,1}| + \sum_{j=2}^{p+2} |b_{p+2,j}| = 0$$

Loss function based on augmented Lagrangian:

$$L_{c,d}(B, \theta, \lambda_1, \lambda_2) = f(B, \theta) + \lambda_1 h_1(B) + \lambda_2 h_2(B) + c|h_1(B)|^2 + d|h_2(B)|^2$$

λ_1, λ_2 Lagrange multiplier; c, d penalty terms

Causal effect decomposition of “Wuhan lockdown” on virus spreading



Simulated Dataset

- Graphs are each 12 nodes
- Varying connection density/sparsity

